Exact one-sided filtering of the two-qubit F_3 functional:
closed forms for both one-sided hidden F_3 measures, every optimal filter, a radical obstruction and success-probability frontiers

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Abstract

Ducuara, Susa and Skrzypczyk defined one-sided hidden correlation measures, in which only Alice or only Bob applies a local filter, and noted that, unlike the two-sided measures, they had no closed formulas (arXiv:2406.03945, Phys. Rev. A **110**, 022435). We give exact formulas for both one-sided F_3 measures of their Equation (10), for every two-qubit state and the invariant functional $F_3(\rho) = \|T\|_F$ of their Equation (3). The squared value is $\beta + \eta + c^T Q(\eta I - Q)^\dagger c$, where (c, Q) are the centre and shape matrix of the other party's steering ellipsoid, η is the larger of $\operatorname{tr} Q - \|c\|^2$ and the largest real eigenvalue of an explicit 6×6 matrix, and $\dagger$ is the Moore–Penrose inverse; a pure filtered marginal gives the product value. The optimum is a supremum, which an invertible filter need not attain. We prove optimality through an explicit S-lemma certificate, describe every optimal filter and decide attainment. A value greater than one has a rational formula and a unique optimal effect. A strictly positive rational state has an optimal squared value of degree six with Galois group S_6, so no formula in radicals exists. Further sections give the exact success-probability frontier, the highest optimal success rate, a sharp bound from the purity of one conditional state and the set of limit values at every pure-marginal state. Lean 4 proves the main theorem for actual complex 4×4 density matrices and invertible complex filters, with a whole-project axiom audit; the extensions and the Galois argument rest on written proofs, exact certificates and numerical checks. The one-sided CHSH measures of Equation (9) are not treated.

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1 Introduction

For a two-qubit state with Pauli correlation matrix T, the Costa–Angelo form of the three-setting linear steering inequality [2, 3] has optimized value $F_3(\rho) = \|T\|_F$, and $F_3 > 1$ certifies steering. Local filtering can reveal such violations when they are absent from the original state. Ducuara, Susa and Skrzypczyk [1] compared the two-sided hidden measures, for which closed expressions are available through the Bell-diagonal normal form (their Equations (7) and (8)), with *one-sided* measures in which only one party filters (their Equations (9) and (10)). They wrote that "there are currently no closed formulas for the one-sided versions defined in (9) and (10)" and listed the derivation of closed formulas among their open questions.

This paper answers that question for both F_3 measures of Equation (10), $\mathcal{H}_A$ and $\mathcal{H}_B$. Theorem 1 gives a finite spectral formula in the steering-ellipsoid data of the other party, valid for every density matrix, with all rank degeneracies and the pure-marginal branch included. Because Equation (10) quantifies invertible filters, its "max" is in general a supremum; we decide exactly when it is attained. The one-sided CHSH measures of Equation (9) remain open. We also note that Equation (8) evaluates F_3 at the normal form; the paper states that it is not known whether this is the two-sided optimum, and we do not address that question.

The question in plain terms. Alice and Bob share two qubits. Before measuring, Alice may pass her qubit through a local filter: a measurement that either succeeds, leaving the pair in a new normalized state, or fails and the round is discarded. A good filter can make the correlations look stronger, so a state that shows no steering may show it after filtering. The question is how large Alice alone can make F_3 in this way, and the same for Bob alone. Naively this is an optimization over all invertible complex 2×2 matrices; we want an explicit answer in terms of the state.

The answer in plain terms. Bob's possible conditional states, as Alice varies her measurement, fill an ellipsoid inside the Bloch ball, his steering ellipsoid [4], with centre c and shape matrix Q. The best value Alice can reach depends only on this ellipsoid. After an explicit change of coordinates on Alice's side, every filter corresponds to a point x of the unit ball, and the squared filtered value becomes a quadratic function of x. Maximizing a quadratic over a ball is a classical problem whose answer is governed by one scalar, a Lagrange multiplier, that solves a polynomial equation. Writing that equation in terms of (c, Q) gives the formula.

Main results.

1. *Closed formula* (Theorem 1). For every state with a mixed filtered marginal, $\mathcal{H}_A^2 = \beta + \eta + c^T Q(\eta I - Q)^\dagger c$, where $\beta = \|c\|^2$ and η is the larger of $\operatorname{tr} Q - \beta$ and the largest real eigenvalue of an explicit 6×6 matrix; a pure filtered marginal gives the product value. Bob's value follows by exchanging the parties. Corollary 2 gives an equivalent rule using six scalar invariants.

2. *Optimal filters and attainment* (Lemma 7, Corollary 8). An explicit gap identity certifies the value, all optimal filters are described, and an invertible optimal filter exists exactly when an explicit condition holds.

3. *Violation is rational* (Theorem 9). Three explicit inequalities decide whether $\mathcal{H}_A > 1$; when it holds, the value has a rational formula and the optimal effect is unique up to scale.

4. *No formula in radicals* (Proposition 10). A strictly positive rational state has an optimal squared value of degree six with Galois group S_6.

5. *Operational extensions* (Sections 11–20). The exact frontier at a prescribed success probability (Theorems 11 and 14), the highest success rate compatible with the optimum (Theorem 16),

continuity (Theorem 21), a sharp bound from the purity of one conditional state (Theorem 25) and the set of limit values at pure-marginal states (Theorem 27).

How the proof works. Lemma 4 computes the filtered correlation matrix for an arbitrary filter and expresses F_3^2 as a ratio of quadratics. Lemma 5 removes the denominator by whitening Alice's marginal; the problem becomes the maximization of a quadratic $q(x)$ over the unit ball. Lemma 6 identifies the multiplier with the largest real eigenvalue of a 6×6 block matrix, including the degenerate "hard case", and Lemma 7 proves optimality by writing the gap between the claimed value and $q(x)$ as a sum of two visibly nonnegative terms. Everything else, including attainment, the rational branch and the filters themselves, is read off from this identity.

Organization. Section 2 fixes definitions and two necessary qualifications. Section 3 states the formula and its variants; Sections 4–9 prove it and derive its consequences, and Section 10 gives examples. Sections 11–20 contains the operational extensions. Appendix A describes the computational checks and Appendix B the Lean formalization.

What is new and what is credited. The spectral solution of a quadratic over a ball (the trust-region subproblem) is classical [8], and two-ball problems have their own literature [9, 10]. The contribution here is the reduction of actual one-sided filtering to such a problem, the closed formula in steering-ellipsoid data, the complete filter and attainment classification, the rational criterion for $\mathcal{H}_A > 1$, the radical obstruction and the operational extensions. Steering ellipsoids and canonical filtering are due to Jevtic, Pusey, Jennings and Rudolph and to Milne et al. [4, 5]; the PPT criterion is used in Corollary 3 [14]. The fixed-centre envelope of F_3 is the subject of a companion package [7]. Steerability under local filters has been classified for assemblages [12], a different object from the invariant optimized here. A bounded literature search in November 2026 found no earlier closed formula for these measures; this does not establish priority.

Formal verification. The Lean development proves, for every positive semidefinite complex 4×4 matrix of trace one, that the explicit right side is the least upper bound of the values of $\|T\|_F$ over all normalized states obtained by invertible complex filters on either side (Appendix B). The rational Bloch-coordinate expressions for (c, Q), the six-invariant rule, the Galois obstruction and the extensions of Sections 11–20 are conventional proofs supported by exact and floating-point checks.

The closed formula

2 Definitions and the two necessary qualifications

Write an arbitrary two-qubit density operator as

$$\rho = \frac{1}{4}\left(I \otimes I + a \cdot \sigma \otimes I + I \otimes b \cdot \sigma + \sum_{i,j=1}^{3} T_{ij}\sigma_i \otimes \sigma_j\right), \quad a, b \in \mathbb{R}^3, \quad T \in \mathbb{R}^{3\times 3}. \tag{1}$$

We use exactly the invariant expression in Equation (3) of [1]: $F_3(\rho) = \sqrt{\mathrm{tr}(T^T T)}$. For $W = A, B$, define

$$\mathcal{H}_W(\rho) = \sup_{K \in \mathrm{GL}(2,\mathbb{C})} F_3\left(\frac{K_W \rho K_W^\dagger}{\mathrm{tr}(K_W \rho K_W^\dagger)}\right), \quad K_A = K \otimes I, \quad K_B = I \otimes K. \tag{2}$$

Multiplying K by a nonzero scalar changes neither the normalized state nor the value, so imposing $K^\dagger K \preceq I$ does not affect the supremum.

Supremum versus maximum. The paper uses "max" while specifying invertible filters. A maximum need not exist. For example, $\rho = \frac{I}{2} \otimes \frac{I+r\sigma_z}{2}$ with $0 < r < 1$ has $\mathcal{H}_A = r$, but every invertible Alice filter gives strictly less than r. All our statements identify exactly when the supremum is attained. Allowing rank-one filters of positive success probability makes it a genuine maximum when the filtered marginal is mixed.

Measurement convention. Optimizing Alice's observables against a permanently fixed Bob Pauli triad gives $\sum_j \|Te_j\| / \sqrt{3}$, generally not $\|T\|_F$. Optimizing the orthonormal triad as well gives $\|T\|_F$. The present theorem concerns the paper's Equation (3) functional, which is the intended invariant input to Equation (10). It does not silently identify these two distinct conventions.

3 The closed spectral formula

For the Alice-filtered quantity assume first $\|a\| < 1$ and set

$$\begin{aligned} \delta &= 1 - \|a\|^2, & B &= T - ab^T, \\ c &= \frac{b - T^T a}{\delta}, & Q &= \frac{B^T(\delta I + aa^T)B}{\delta^2}. \end{aligned} \tag{3}$$

Here (c, Q) is Bob's steering-ellipsoid centre and shape matrix; $Q \succeq 0$ may be singular. These expressions are rational in the original Bloch data. Define

$$\tau = \operatorname{tr} Q, \qquad \beta = \|c\|^2, \qquad d = \tau - \beta, \qquad \mathscr{M}(Q, c) = \begin{pmatrix} Q & I_3 \\ Qcc^T & Q \end{pmatrix}. \tag{4}$$

The matrix $\mathscr{M}$ need not be symmetric. In the following, rmax means its *largest real eigenvalue*, not its spectral radius or largest eigenvalue magnitude. Such a real eigenvalue always exists here.

Theorem 1 (Exact one-sided filtering)**.** *With (3)–(4), put*

$$\nu = \operatorname{rmax} \mathscr{M}(Q, c), \qquad \eta = \max\{d, \nu\}, \qquad D = \eta I_3 - Q. \tag{5}$$

Then

$$\boxed{\mathcal{H}_A(\rho)^2 = \beta + \eta + c^T Q(\eta I_3 - Q)^\dagger c.} \tag{6}$$

The dagger denotes the Moore–Penrose inverse, including at zero or repeated eigenvalues. For Bob, use precisely the same formula after the substitution

$$(a, b, T) \longmapsto (b, a, T^T). \tag{7}$$

If the filtered marginal is pure, the corresponding formula instead is

$$\|a\| = 1 \implies \mathcal{H}_A(\rho) = \|b\|, \qquad \|b\| = 1 \implies \mathcal{H}_B(\rho) = \|a\|. \tag{8}$$

The pure-marginal values are attained by every invertible filter on that side.

This is a finite algebraic spectral expression, analogous in its use of matrix spectra to other exact two-qubit criteria. It is stronger than a reformulation as an optimization problem. Section 9 proves that requiring radicals instead would be impossible in general.

An equivalent explicit polynomial version is often convenient. Diagonalize Q with eigenvalues $\kappa_1, \kappa_2, \kappa_3 \geq 0$, and write c_i for the centre components in that orthonormal eigenbasis. Then ν is the largest real root of

$$P(z) = \prod_{i=1}^{3} (z - \kappa_i)^2 - \sum_{i=1}^{3} \kappa_i c_i^2 \prod_{j \neq i} (z - \kappa_j)^2. \tag{9}$$

Repeated eigenvalues and zero κ_i are allowed; no cancellation of factors is to be performed before choosing the largest root. Finally,

$$\mathcal{H}_A^2 = \beta + \eta + \sum_{\kappa_i < \eta} \frac{\kappa_i c_i^2}{\eta - \kappa_i}. \tag{10}$$

For every omitted index with $\kappa_i = \eta$, the construction guarantees $\sqrt{\kappa_i} c_i = 0$. Thus the omitted term means zero, not $0/0$.

3.1 A formula using six scalar invariants only

The pseudoinverse and eigenbasis can both be eliminated from evaluation of the value. This also gives an efficient invariant representation of the sextic. Set

$$s_2 = \frac{\tau^2 - \operatorname{tr}(Q^2)}{2}, \quad s_3 = \det Q, \quad u = c^T Q c, \quad v = c^T Q^2 c,$$

and retain $\tau = \operatorname{tr} Q$, $\beta = c^T c$. Define the cubic and quadratic

$$D(z) = z^3 - \tau z^2 + s_2 z - s_3, \qquad N(z) = u z^2 + (v - \tau u) z + s_3 \beta. \tag{11}$$

Corollary 2 (Six-invariant algebraic formula, including poles)**.** *The characteristic sextic in (9) is exactly*

$$\boxed{P(z) = D(z)^2 + D(z) N'(z) - D'(z) N(z).} \tag{12}$$

Let ν be its largest real root and $\eta = \max\{\tau - \beta, \nu\}$. Let r be the first index in $\{0, 1, 2, 3\}$ with $D^{(r)}(\eta) \neq 0$, where $D^{(0)} = D$. Then

$$\boxed{\mathcal{H}_A(\rho)^2 = \beta + \eta + \frac{N^{(r)}(\eta)}{D^{(r)}(\eta)}.} \tag{13}$$

Thus the six real numbers $(\tau, s_2, s_3, \beta, u, v)$ determine the exact value. No matrix diagonalization, pseudoinverse, limiting operation, continuous optimization or pole cancellation is needed to evaluate this rule. The Bob version uses its own swapped ellipsoid data.

Proof. Cayley–Hamilton gives

$$\operatorname{adj}(zI - Q) = z^2 I + z(Q - \tau I) + Q^2 - \tau Q + s_2 I, \qquad Q^3 - \tau Q^2 + s_2 Q = s_3 I.$$

Hence $D(z) = \det(zI - Q)$ and $N(z) = c^T Q \operatorname{adj}(zI - Q) c$. Away from the spectrum, $R(z) = N(z)/D(z) = c^T Q (zI - Q)^{-1} c$ has $R'(z) = -c^T Q (zI - Q)^{-2} c$. The determinant lemma for the companion matrix therefore gives $P(z) = D(z)^2 (1 + R'(z))$, which expands to (12). Equality away from finitely many points is polynomial equality everywhere.

If $D(\eta) \neq 0$, the value correction is simply $N(\eta)/D(\eta)$. Otherwise, the main theorem's compatibility condition annihilates the component of $Q^{1/2} c$ in the η-eigenspace. Thus R has a removable singularity at η, and its value there is exactly $c^T Q (\eta I - Q)^\dagger c$. If r is the multiplicity of η in D, divisibility makes the first r powers cancel in both N and D; the ratio of their coefficients at that order is $N^{(r)}(\eta)/D^{(r)}(\eta)$. The factorials cancel. This covers $r = 3$ as well: compatibility then makes N identically zero. Only the finite derivative rule is used in (13). □

Corollary 3 (Radicals for every rank-deficient ellipsoid)**.** *If $\operatorname{rank} Q \leq 2$, the original state is separable, so both one-sided values are at most one. The exact value is expressible by radicals: P has a factor z^2, leaving at most a quartic. A uniform root-selection rule retains zero as a candidate; at rank exactly two the largest real root is positive and comes from the quartic. If $\operatorname{rank} Q \leq 1$, the formula simplifies to*

$$\boxed{\mathcal{H}_A^2 = \beta + \tau + 2\sqrt{c^T Q c} \leq 1.} \tag{14}$$

For a mixed filtered marginal in this rank-at-most-one case, the raw F_3 optimum is attained invertibly if and only if $c = 0$. These ranks are ranks of the ellipsoid shape Q, not of the density matrix.

Proof. For the separability statement, use the canonical state $(0, c, S)$. If $\operatorname{rank} S \leq 2$, an Alice rotation makes the third row of S zero. Alice's partial transpose flips its second row; conjugation by $\sigma_x \otimes I$ flips both the second and third rows and therefore produces the same state. The partial transpose is consequently positive. The two-qubit PPT criterion [14] makes the canonical state separable, and inverse local whitening preserves separability. Local filters then preserve separability, and convexity of the Frobenius norm of product correlation matrices gives both one-sided bounds by one.

Rank at most two gives $s_3 = 0$ in (11); substitution into (12) shows the factor z^2 directly. At rank two, $\nu \geq \lambda_{\max}(Q) > 0$. The value formula then uses rational operations and finite derivatives of a degree-at-most-four algebraic number, followed by a square root for $\mathcal{H}_A$.

At rank at most one, $s_2 = s_3 = 0$ and $v = \tau u$, so $P(z) = z^4[(z-\tau)^2 - u]$. Hence $\nu = \tau + \sqrt{u}$ and $\eta = \nu$, since $d = \tau - \beta \leq \tau$. Substitution, including the removable-pole convention when $u = 0$, gives (14). For $Q = \tau e e^T$ its right side is $\max\{\|c + \sqrt{\tau}e\|^2, \|c - \sqrt{\tau}e\|^2\}$. Both endpoints are physical conditional Bloch vectors, proving the upper bound one; $Q = 0$ is the same statement with coincident endpoints. The multiplier is $\lambda = \eta - d = \beta + \sqrt{u}$, so an invertible optimizer requires $\beta = 0$. Conversely $c = 0$ gives $g = 0$, $\lambda = 0$ and the interior optimum $x = 0$. □

4 Bloch geometry and elimination of the fractional denominator

Lemma 4 (Fractional quadratic identity)**.** *For an Alice filter whose effect is proportional to $I + x \cdot \sigma$, with $\|x\| < 1$, set $t = \|T\|_F^2$, $u = Tb$, and $H_0 = TT^T + (\|b\|^2 - t)I_3$. Then*

$$F_3(\rho_{A,x})^2 = \frac{t + 2u^T x + x^T H_0 x}{(1 + a^T x)^2}. \tag{15}$$

For $\|a\| < 1$ the expression extends continuously to the closed unit ball.

Proof. Polar decomposition removes the output unitary, since it only rotates T. After removing an irrelevant positive scalar, the filter is $\sqrt{I + x \cdot \sigma}$. Put $s = \sqrt{1 - \|x\|^2}$ and

$$L_x = sI_3 + \frac{xx^T}{1+s}.$$

The Pauli identity $(v \cdot \sigma)(w \cdot \sigma) = (v \cdot w)I + i(v \times w) \cdot \sigma$ gives the transformed correlation matrix

$$T_x = \frac{L_x T + xb^T}{1 + a^T x}.$$

Since $L_x x = x$ and $L_x^2 = (1 - \|x\|^2)I + xx^T$, its squared Frobenius norm has numerator

$$(1 - \|x\|^2)t + x^T TT^T x + 2x^T Tb + \|x\|^2 \|b\|^2,$$

which is (15). The denominator is at least $(1 - \|a\|)^2 > 0$ on the closed ball. □

Lemma 5 (Whitening and canonical data)**.** *For $\|a\| < 1$, let*

$$W = (I + a \cdot \sigma)^{-1/2}, \qquad L_a = \sqrt{\delta} I_3 + \frac{aa^T}{1 + \sqrt{\delta}}, \tag{16}$$

$$S = \frac{L_a T - ab^T}{\delta}.$$

Then $(W \otimes I)\rho(W \otimes I)$ already has trace one and Bloch data $(0, c, S)$. Moreover $S^T S = Q$. Its one-sided filtering supremum equals the original state's supremum.

Proof. The Alice marginal becomes $W(I + a \cdot \sigma)W/2 = I/2$, proving the trace and zero-marginal claims. Applying Lemma 4's Pauli transformation with $x = -a$ gives c and S. Since $L_a a = a$, $S = L_a(T - ab^T)/\delta$, and $L_a^2 = \delta I + aa^T$ proves $S^T S = Q$. Multiplication by the invertible matrix W is a bijection on all invertible filter matrices, up to their irrelevant scalars. Thus it preserves the optimization orbit, even though W itself may need rescaling to be a valid measurement operator. □

For the canonical state define

$$C = SS^T, \qquad g = Sc, \qquad q(x) = \tau + 2g^T x + x^T(C - dI)x. \tag{17}$$

The remaining task is exactly $\max_{\|x\| \le 1} q(x)$; its value is also the supremum over the open ball by continuity. Notice the special boundary identity

$$\|x\| = 1 \quad \Longrightarrow \quad q(x) = \left\| c + S^T x \right\|^2 \le 1. \tag{18}$$

The equality follows by expanding both sides. The inequality follows either from the steering ellipsoid lying in the Bloch ball or from the fact that a rank-one Alice filter produces a product state with a pure Alice factor.

5 Exact spectral solution, including the hard case

We prove the spectral step explicitly instead of appealing to an SDP solver. In this section write $\kappa = \lambda_{\max}(C)$ and use an orthonormal eigenbasis of the real symmetric C. Let g_i be the components of g. For $z > \kappa$ define

$$f(z) = \sum_i \frac{g_i^2}{(z - \kappa_i)^2}. \tag{19}$$

If $g \neq 0$, f is continuous, strictly decreasing, and tends to zero at infinity. At the top eigenvalue it tends to infinity if g has a component in the top eigenspace. Otherwise its one-sided limit is

$$f(\kappa+) = \sum_{\kappa_i < \kappa} \frac{g_i^2}{(\kappa - \kappa_i)^2}.$$

Lemma 6 (Spectral parameter)**.** *Let*

$$M_C = \begin{pmatrix} C & I \\ gg^T & C \end{pmatrix}.$$

Its largest real eigenvalue ν exists and satisfies $\nu \ge \kappa$. Exactly one of the following descriptions applies:

1. *If $f(\kappa+) > 1$, $\nu > \kappa$ is the unique solution of $f(\nu) = 1$.*
2. *Otherwise $\nu = \kappa$, g is orthogonal to the top eigenspace, and $\left\|(\nu I - C)^\dagger g\right\| \le 1$.*

The convention $f \equiv 0$ covers $g = 0$.

Proof. Block elimination gives the polynomial identity

$$\det(zI_6 - M_C) = \det((zI_3 - C)^2 - gg^T). \tag{20}$$

For $z > \kappa$, the determinant lemma makes this $\det(zI - C)^2(1 - f(z))$. The intermediate value theorem and strict monotonicity prove the first case and exclude any larger real root. In the second case g has no top-eigenspace component. A top eigenvector is therefore in the kernel of $(\kappa I - C)^2 - gg^T$, so κ is a root. Monotonicity excludes roots above it. The stated norm is exactly the limit $f(\kappa+)$ under a square root. This also treats repeated top eigenvalues. □

Set $\eta = \max(d, \nu)$, $A = \eta I - C$, $\lambda = \eta - d$, and $x_0 = A^\dagger g$. The lemma proves

$$\lambda \geq 0, \quad A \succeq 0, \quad Ax_0 = g, \quad \|x_0\| \leq 1. \tag{21}$$

If $\eta > \nu$, A is positive definite, $\lambda = 0$, and $\|x_0\| < 1$. If $\eta = \nu > \kappa$, the norm is one. If $\eta = \nu = \kappa$, the kernel of A is nonzero; one may add a kernel vector to fill any missing norm. Consequently there always exists

$$x_* = x_0 + z, \quad z \in \ker A, \quad \|x_*\| \leq 1, \quad \lambda(1 - \|x_*\|^2) = 0. \tag{22}$$

When $\lambda > 0$ and the norm of x_0 is less than one, one can choose any unit $e \in \ker A$ and take $z = \sqrt{1 - \|x_0\|^2}\, e$. The range of a symmetric matrix is orthogonal to its kernel, so this construction is exact.

Lemma 7 (Gap identity and complete equality classification)**.** *Let* $V = \beta + \eta + g^T A^\dagger g = \tau + \lambda + g^T x_0$. *For every* x,

$$\boxed{V - q(x) = \lambda(1 - \|x\|^2) + (x - x_0)^T A (x - x_0).} \tag{23}$$

Thus V *is the global maximum on the unit ball. All maximizing points are exactly*

$$x = x_0 + z, \quad z \in \ker A, \quad \|x\| \leq 1, \quad \lambda(1 - \|x\|^2) = 0. \tag{24}$$

Proof. Expand the right side using $Ax_0 = g$ and $A = \lambda I - (C - dI)$. Both summands are nonnegative on the ball. Construction (22) makes both zero. Conversely, equality forces both nonnegative terms to vanish, and a positive semidefinite quadratic form vanishes exactly on its kernel. This proves the full classification, including all singular cases. □

To finish the spectral part of Theorem 1, choose an orthogonal extension of the polar decomposition $S = O\sqrt{Q}$. Such an extension exists even when S is singular. Under $x \mapsto O^T x$, (C, g) becomes $(Q, \sqrt{Q}c)$. Thus $g^T(\eta I - C)^\dagger g = c^T Q(\eta I - Q)^\dagger c$. The characteristic polynomial of the resulting symmetric-Gram block matrix equals that of $\mathscr{M}(Q, c)$: away from the eigenvalues of Q, both are

$$\det(zI - Q)^2 \left(1 - c^T Q (zI - Q)^{-2} c\right).$$

Since these are polynomials, equality on that open set is equality everywhere. This proves (6) and (9) without assuming Q invertible.

6 The S-lemma certificate is explicit

The ball has a strictly feasible point. The S-lemma therefore identifies upper bounds on the quadratic as nonnegative quadratic certificates. Here the certificate is supplied in closed form, with no remaining multiplier optimization. In canonical coordinates,

$$\begin{pmatrix} V - \tau - \lambda & -g^T \\ -g & \lambda I - (C - dI) \end{pmatrix} = \begin{pmatrix} x_0^T A x_0 & -x_0^T A \\ -Ax_0 & A \end{pmatrix} \succeq 0. \tag{25}$$

It factors as $\binom{x_0^T}{-I} A \begin{pmatrix} x_0 & -I \end{pmatrix}$. Together with $\lambda \geq 0$, this is exactly the identity (23). The separately constructed maximizing point proves there is no duality gap.

For comparison, applying the S-lemma directly to the fractional expression (15) says that v is a global upper bound if and only if there exists $s \geq 0$ such that

$$\begin{pmatrix} v - t - s & (va - u)^T \\ va - u & vaa^T - H_0 + sI \end{pmatrix} \succeq 0. \tag{26}$$

This display alone would only be an SDP restatement. Whitening followed by (5)–(6) eliminates the unknown optimization variables and provides the exact value and an explicit saturation certificate.

7 Every extremizing filter and every singular case

For $\|a\| < 1$, compute S as in (16), and put

$$A = \eta I - SS^T, \qquad x_0 = A^\dagger Sc, \qquad \lambda = \eta - d. \tag{27}$$

Choose any x in (24). Every corresponding maximizer in the closed filter domain, including rank-one limiting filters, is, up to the allowed equivalences,

$$\boxed{K = \gamma U \sqrt{I + x \cdot \sigma}\,(I + a \cdot \sigma)^{-1/2},} \tag{28}$$

where U is an arbitrary qubit unitary and $\gamma > 0$ is any scaling that makes $K^\dagger K \preceq I$. Conversely, polar decomposition after whitening shows that every optimal filter is of this form.

For a direct description in the original normalized effect coordinates, the same filter has effect proportional to $I + y \cdot \sigma$, where

$$y = \frac{L_a x - a}{1 - a \cdot x}, \qquad 1 - \|y\|^2 = \frac{\delta(1 - \|x\|^2)}{(1 - a \cdot x)^2}. \tag{29}$$

Indeed $W(I + x \cdot \sigma)W = [(1 - a \cdot x)I + (L_a x - a) \cdot \sigma]/\delta$. This is a ball automorphism; it explicitly relates the fractional quadratic coordinates to the canonical quadratic coordinates and preserves the interior/boundary distinction.

Corollary 8 (Attainment by invertible filters)**.** *For a mixed filtered marginal, the supremum in Equation (10) is attained by an invertible filter if and only if*

$$\lambda = 0 \quad \textit{and} \quad \|x_0\| < 1. \tag{30}$$

Every invertible maximizer uses precisely the points of (24) *with norm strictly less than one. Otherwise all maximizing points have norm one, and replacing such a point by* $(1 - \varepsilon)x$ *gives an explicit sequence of invertible filters approaching the supremum.*

Proof. The positive matrix $I + x \cdot \sigma$ is invertible exactly for $\|x\| < 1$. Since $x_0 \perp \ker A$, it is the minimum-norm member of $x_0 + \ker A$. The claimed condition follows from (24). □

The success probability is also explicit. With K_0 denoting (28) before multiplying by γ, the canonical trace normalization gives $\mathrm{tr}(K_0 \rho_A K_0^\dagger) = 1$. Therefore the maximum success probability among scalar multiples of this effect is

$$\begin{aligned} p_{\max}(x) &= \frac{1}{\lambda_{\max}(W(I + x \cdot \sigma)W)} \\ &= \frac{\delta}{1 - a \cdot x + \sqrt{(1 - a \cdot x)^2 - \delta(1 - \|x\|^2)}}. \end{aligned} \tag{31}$$

This follows from the trace and determinant of the positive 2×2 matrix in the first line. The denominator is positive for $\|a\| < 1$. In particular a boundary optimum here has a nonzero limiting success probability; nonattainment under the GL convention need not mean a zero-probability limit.

Pure marginal. If $\|a\| = 1$, rotate Alice so her marginal is $|0\rangle\langle 0|$. In the corresponding block form of ρ, the lower diagonal positive block has trace zero and is zero. Positivity forces its off-diagonal blocks to vanish (equivalently apply Cauchy–Schwarz to the positive form). Thus $\rho = |0\rangle\langle 0| \otimes \rho_B$. Every nonzero successful Alice filter preserves the product structure and the purity of her factor. Its correlation matrix has squared norm $\|b\|^2$, proving (8). A rank-one filter annihilating the occupied Alice state has zero success probability and is excluded.

Other rank deficiencies. No assumption on $\operatorname{rank}\rho$, $\operatorname{rank}T$, or $\operatorname{rank}Q$ entered the mixed-marginal proof. A singular A uses its pseudoinverse and the explicitly described kernel freedom. Repeated eigenvalues do not require a preferred eigenbasis. The cases $Q=0$, $g=0$, and a zero multiplier are all included. In particular, if $Q=0$ the value is $\|c\|$; if $c=0$ the squared value is simply $\operatorname{tr}Q$.

8 The entire violating branch is rational

Theorem 9 (Rational criterion and unique violating optimizer)**.** *For a mixed filtered marginal, put $D_0 = dI - Q$. Then $\mathcal{H}_A > 1$ if and only if all three conditions hold:*

$$\begin{aligned} D_0 &\succ 0, \\ c^T Q D_0^{-2} c &< 1, \\ \tau + c^T Q D_0^{-1} c &> 1. \end{aligned} \tag{32}$$

Whenever they hold,

$$\boxed{\mathcal{H}_A^2 = \tau + c^T Q (dI - Q)^{-1} c, \qquad x_* = (dI - SS^T)^{-1} S c.} \tag{33}$$

The optimal filter effect is unique up to positive scalar, and all optimal filters differ only by that scalar and an output unitary. For rational original Bloch data, the squared optimum on this branch is rational.

Proof. Suppose the maximum exceeds one. By (18), every optimizer is interior. The quadratic's stationary equation holds and its Hessian $2(C - dI)$ is negative semidefinite. If it had a null vector, stationarity would make the quadratic constant on a line through the optimum in that direction; that line reaches the sphere, contradicting (18). Hence $dI - C \succ 0$. The unique stationary point is $(dI - C)^{-1} g$, has norm less than one, and yields $\tau + g^T (dI - C)^{-1} g$. Conjugating by the polar orthogonal map proves all three conditions and (33). Conversely these conditions describe a strictly concave quadratic with its unique stationary point strictly inside the ball and value greater than one. It is the global maximum. Rationality follows from (3) and the ordinary matrix inverse formula. □

Thus high-degree algebra is needed only for nonviolating values, not for deciding whether the one-sided filtered functional can exceed one. In particular, if $\kappa_1 \geq \kappa_2 \geq \kappa_3$ are the eigenvalues of Q, a useful geometric exclusion criterion is

$$\|c\|^2 \geq \kappa_2 + \kappa_3 \quad \Longrightarrow \quad \mathcal{H}_A \leq 1. \tag{34}$$

Indeed $D_0 \succ 0$ would require $d > \kappa_1$, equivalently $\|c\|^2 < \kappa_2 + \kappa_3$. The full criterion (32), rather than this necessary geometric condition alone, decides violation.

9 Why a universal radical formula is impossible

Consider the rational canonical Bloch data

$$a = 0, \qquad c = (1/5, 1/6, 1/7)^T, \qquad S = \operatorname{diag}(1/20, 1/15, 1/10). \tag{35}$$

These describe a strictly positive density matrix: $\|c\| < 1/3$ and $\sum_i |S_{ii}| = 13/60$, so the norm of the nonidentity part in (1) is less than $11/20 < 1$. In fact every density eigenvalue is greater than $9/80$. Here $\tau = 61/3600 < \beta = 3889/44100$, so $C - dI$ is positive definite and the optimum is a boundary optimum.

The primitive integer polynomial obtained from (9) is

$$P_0(z) = 3175200000000000z^6 - 107604000000000z^5 + 65685000000z^4 + 18599100000z^3 - 188411500z^2 + 695840z - 913. \tag{36}$$

The included standard-library-only checker reconstructs these coefficients from exact fractions and verifies these squarefree modular factorizations:

$$P_0 \equiv 5(z+6)(z^5+16z^4+4z^3+13z^2+9z+3) \pmod{17},$$
$$P_0 \equiv 22(z^6+55z^5+2z^4+35z^3+2z^2+48z+47) \pmod{59},$$
$$P_0 \equiv 188(z+460)(z+355)(z+332)(z+114)(z^2+309z+411) \pmod{487}.$$

It proves the degree-five, degree-six, and degree-two displayed factors irreducible using finite-field polynomial arithmetic, not a numerical factorization or a call to a symbolic Galois-group routine.

Proposition 10 (Physical obstruction to radicals)**.** *For (35), neither $\mathcal{H}_A^2$ nor $\mathcal{H}_A$ is expressible by radicals over $\mathbb{Q}$. Consequently no formula using only rational operations and radicals in arbitrary rational Bloch data can give both one-sided filtered measures in general.*

Proof. Irreducibility modulo 59 implies irreducibility over $\mathbb{Q}$ and a transitive degree-six Galois group. The squarefree factor patterns modulo 17 and 487 give respectively a 5-cycle and a transposition. A transitive group of degree six containing a 5-cycle is primitive: a nontrivial block system would have blocks of size two or three, and the 5-cycle could neither move the fewer-than-five blocks nor act nontrivially inside a block of size less than five. A primitive group containing a transposition is S_6. To see this directly, form the graph whose edges are all conjugates of that transposition. Its connected components form a block system; primitivity makes the graph connected, and edge transpositions of a connected graph generate the full symmetric group. Therefore the normal closure has group S_6.

Let ν be the largest real root of P_0 and $K = \mathbb{Q}(\nu)$. The optimum is

$$V = \beta + \nu + \sum_i \frac{\kappa_i c_i^2}{\nu - \kappa_i} \in K.$$

It is not rational: after multiplying by $\prod_i(\nu - \kappa_i)$, an alleged rational value would give a nonzero degree-four polynomial, with leading coefficient one, annihilating a degree-six algebraic number. In the S_6 normal closure the field K is fixed by a point stabilizer S_5, a maximal subgroup. Thus K has no proper intermediate fields and $\mathbb{Q}(V) = K$. Its nonsolvable normal closure precludes expression of V by radicals. Expression of $\sqrt{V}$ by radicals would imply one for V and is also impossible. □

The exact checker also isolates $0.02806139 < \nu < 0.02806140$ and encloses the optimum using rational interval endpoints; numerically $V \simeq 0.1366862793$. These decimal approximations are illustrations, not inputs to the Galois proof.

10 Examples that exercise different branches

Werner states. For $a = b = 0$ and $T = -pI$, $0 \le p \le 1$, one obtains $\mathcal{H}_A^2 = \mathcal{H}_B^2 = 3p^2$. The identity filter is optimal. The case $p = 0$ is entirely flat.

A full-rank example without an invertible maximizer. For $a = 0$, $T = 0$, $b = (0, 0, r)$, $0 < r < 1$, one has $Q = 0$, $\nu = \eta = 0$, $\mathcal{H}_A^2 = r^2$. Here $q(x) = r^2 \|x\|^2$: exactly the sphere maximizes. This disproves literal universal existence of the maximum written in Equation (10) under a GL-only convention.

A singular flat optimum. For $a = b = 0$, $T = \mathrm{diag}(0, 0, s)$ with $|s| \leq 1$, the quadratic is $q(x) = s^2(1 - x_1^2 - x_2^2)$. Every $(0, 0, x_3)$ with $|x_3| \leq 1$ is optimal. Both invertible and rank-one optimal filters occur. An ordinary inverse formula would fail, while the pseudoinverse formula is exact.

A nonzero-gradient hard case. The physical canonical data

$$c = (0, 0, 3/10), \qquad S = \mathrm{diag}(1/5, 1/10, 1/50)$$

yield a positive multiplier, a singular optimal matrix, and a nonzero gradient. They give $\mathcal{H}_A^2 = 36/275 \simeq 0.1309090909$. The pseudoinverse solution must be completed in the top eigenspace to have norm one. This checks a hard case that does not reduce to $g = 0$.

Both directions exhibit different exact hidden values. Consider the amplitude-damping Choi state with

$$a = 0, \quad b = (0, 0, 3/5), \quad T = \mathrm{diag}(\sqrt{2/5}, -\sqrt{2/5}, 2/5).$$

Its unfiltered $F_3^2 = 24/25 < 1$, whereas

$$\mathcal{H}_A^2 = \frac{12}{11}, \qquad \mathcal{H}_B^2 = \frac{120}{71}. \tag{37}$$

For Alice the optimal canonical vector is $(0, 0, 6/11)$ and the maximum success probability is $11/17$. For Bob it is $(0, 0, -15/71)$ after whitening Bob; the maximum success probability is $71/215$. Bob's canonical data are $c = (0, 0, -3/8)$ and $Q = \mathrm{diag}(5/8, 5/8, 25/64)$, which directly reproduce the second fraction.

Operational and geometric extensions

11 Prescribed success probability: an exact two-ball geometry

The unrestricted quantity ignores how frequently the chosen outcome occurs. Filtering-probability tradeoffs have been studied for particular systems [13]. We now obtain an arbitrary-state one-sided formula. For $0 < p \le 1$ define

$$\mathcal{H}_A(\rho; p) = \sup\left\{F_3(\rho_K) : K^\dagger K \preceq I,\ \mathrm{tr}[(K^\dagger K \otimes I)\rho] \ge p\right\}, \qquad V(p) = \mathcal{H}_A(\rho; p)^2.$$

The supremum may be restricted to invertible filters without changing its value. We prove that assertion rather than assume attainment. All the results in this part have a Bob version by the same swap as in (7). We retain the canonical notation

$$C = SS^T, \quad g = Sc, \quad d = \mathrm{tr}\, C - \|c\|^2, \quad H = C - dI, \quad q(x) = \tau + 2g^T x + x^T H x.$$

Theorem 11 (Exact probability lens)**.** *If $\|a\| < 1$, then*

$$\boxed{V(p) = \max_{x \in L_p} q(x), \qquad L_p = \{x : \|x\| \le 1,\ \|a - px\| \le 1 - p\}.} \tag{38}$$

For $0 < p < 1$ this is the intersection of the two explicit balls $B(0,1)$ and $B(a/p, (1-p)/p)$. Each feasible point is realized with success probability exactly p by

$$K = \sqrt{p}\, U \sqrt{I + x \cdot \sigma}\, W. \tag{39}$$

At $p = 1$, $L_p = \{a\}$ and $V(1) = F_3(\rho)^2$. If $\|a\| = 1$, the entire frontier is instead constant: $\mathcal{H}_A(\rho; p) = \|b\|$ for every $0 < p \le 1$.

Proof. An admissible filter of probability $P \ge p$ can be multiplied by $\sqrt{p/P}$ without changing its output. Thus it suffices to prescribe probability exactly p. The canonical marginal is $I/2$, so (39) has that probability. Positivity of its canonical effect is exactly $\|x\| \le 1$. Moreover

$$K^\dagger K \preceq I \iff p(I + x \cdot \sigma) \preceq I + a \cdot \sigma \iff \|a - px\| \le 1 - p.$$

Polar decomposition proves that this parametrization covers every output. The lens is compact and nonempty, since it contains a.

For $p < 1$, a is strictly inside both balls: $\|a\| < 1$ and $\|a - pa\| = (1-p)\|a\| < 1 - p$. Consequently $x_\varepsilon = (1-\varepsilon)x + \varepsilon a$ remains feasible and has $\|x_\varepsilon\| < 1$. Continuity of q shows that the closed-lens maximum equals the invertible-filter supremum. At $p = 1$, the only point is a, and $\sqrt{I + a \cdot \sigma} W = I$. The pure-marginal statement follows from the product-state reduction already proved. □

Corollary 12 (Containment regimes and a universal achievable rate)**.** *Put $r_a = \|a\|$, $p_- = (1 - r_a)/2$ and $p_+ = (1 + r_a)/2$. For $p \le p_-$ the lens is the full unit ball. For $p_+ \le p < 1$ it is the second ball. Only $p_- < p < p_+$ requires a genuine lens. Every unrestricted optimum in the closed filter domain therefore has some realization with success at least p_-. Every strictly violating optimum has a realization with success strictly greater than p_-.*

Proof. The two containments are respectively equivalent to $r_a/p + 1 \le (1-p)/p$ and $r_a/p + (1-p)/p \le 1$. For the strict assertion use the unique interior optimizer and

$$W(I + x \cdot \sigma)W \preceq \frac{1 + \|x\|}{1 - \|a\|} I, \qquad p_{\max}(x) \ge \frac{1 - \|a\|}{1 + \|x\|} > p_-.$$

□

Corollary 13 (Canonical frontier: a single six-dimensional spectrum)**.** *If $a = 0$ and $0 < p < 1$, put $R = \min\{1, (1-p)/p\} > 0$. Define*

$$\nu_R = \operatorname{rmax} \begin{pmatrix} Q & I \\ Qcc^T/R^2 & Q \end{pmatrix}, \qquad \eta_R = \max\{d, \nu_R\}.$$

Then

$$\boxed{V(p) = \tau + R^2(\eta_R - d) + c^T Q(\eta_R I - Q)^\dagger c.} \tag{40}$$

In particular the unrestricted canonical optimum is achievable with probability at least $1/2$, and a strictly violating optimum with probability strictly greater than $1/2$.

Proof. The two balls are concentric. Apply the quadratic theorem to $x = Ry$. Equivalently, the regular secular equation is $\left\|(\eta_R I - C)^{-1} g\right\|^2 = R^2$, with the same kernel completion in the hard case. For $x_0 = (\eta_R I - C)^\dagger g$ the exact identity is

$$V(p) - q(x) = (\eta_R - d)(R^2 - \|x\|^2) + (x - x_0)^T(\eta_R I - C)(x - x_0).$$

All maximizers follow by setting its nonnegative terms to zero. □

For completeness, the translated-ball branch is equally explicit. On $B(u, R)$, $R > 0$, set $g_u = g + Hu$, $\tau_u = q(u)$ and

$$\nu_u = \operatorname{rmax} \begin{pmatrix} H & I \\ g_u g_u^T/R^2 & H \end{pmatrix}, \qquad \mu_u = \max\{0, \nu_u\}.$$

Its maximum is

$$\tau_u + \mu_u R^2 + g_u^T(\mu_u I - H)^\dagger g_u. \tag{41}$$

Here $u = a/p$, $R = (1-p)/p$. A zero radius is the single-point case.

12 A finite algebraic formula for the genuine lens

Generic quadratic optimization over two balls is established mathematics [9, 10]. The contribution of this section is the explicit fixed-dimensional candidate formula for the physical domain (38), including feasible singular families and all relevant stationary roots. It does not assume that a two-constraint S-lemma is exact.

Theorem 14 (At most fourteen candidate values)**.** *Let $q(x) = \tau + 2g^T x + x^T H x$ with $H = H^T \in \mathbb{R}^{3\times 3}$. In the transverse lens regime of Corollary 12, its maximum is the largest of the following explicitly constructed feasible values: one stationary-affine value, at most six stationary values from each boundary sphere, and one common-circle value. Every feasibility test below uses only arithmetic, square roots, matrix spectra and pseudoinverses. There is no remaining continuous optimization.*

Candidate 1: the stationary affine family. Write the two ball centres and radii as (u_0, R_0) and (u_1, R_1). For (38) these are $(0, 1)$ and $(a/p, (1-p)/p)$. If $g \notin \operatorname{Ran} H$, there is no candidate of this type. Otherwise set

$$x_s = -H^\dagger g, \qquad K = \ker H, \qquad V_s = \tau - g^T H^\dagger g,$$

and let P_K be orthogonal projection onto K. Define

$$k_i = P_K(u_i - x_s), \qquad s_i^2 = R_i^2 - \|(I - P_K)(x_s - u_i)\|^2.$$

Include V_s exactly when $s_0^2, s_1^2 \geq 0$ and

$$\|k_0 - k_1\| \leq \sqrt{s_0^2} + \sqrt{s_1^2}. \tag{42}$$

A witness is $x_s + z$ with $z = k_0$ if $k_0 = k_1$; otherwise use

$$z = k_0 + \min\left\{1, \frac{\sqrt{s_0^2}}{\|k_1 - k_0\|}\right\}(k_1 - k_0).$$

This remains valid when $K = \{0\}$.

Candidates 2–13: all stationary sphere families. For $i \in \{0, 1\}$ put $g_i = g + Hu_i$ and $\tau_i = q(u_i)$. Choose an orthonormal eigenbasis of H with eigenvalues h_j and components g_{ij} of g_i. Form the degree-six polynomial

$$P_i(\ell) = R_i^2 \prod_{j=1}^{3}(\ell - h_j)^2 - \sum_{j=1}^{3} g_{ij}^2 \prod_{k \neq j}(\ell - h_k)^2. \tag{43}$$

The roots also are the eigenvalues of $\left(\begin{smallmatrix} H & I \\ g_i g_i^T/R_i^2 & H \end{smallmatrix}\right)$. Every *real* root must be considered, including roots below the largest real root.

If $\ell \notin \operatorname{spec} H$, set

$$y = (\ell I - H)^{-1} g_i, \qquad V_i(\ell) = \tau_i + \ell R_i^2 + g_i^T y.$$

Include this value exactly when $\|u_i + y - u_j\| \leq R_j$, $j \neq i$. The root equation already gives $\|y\| = R_i$.

For a spectral value ℓ, put $K_\ell = \ker(\ell I - H)$. Require $g_i \perp K_\ell$ and $y_0 = (\ell I - H)^\dagger g_i$ with $\|y_0\| \leq R_i$. Set $s = \sqrt{R_i^2 - \|y_0\|^2}$ and let P_ℓ project onto K_ℓ. The family and its constant value are

$$x = u_i + y_0 + z, \quad z \in K_\ell, \quad \|z\| = s, \qquad V_i(\ell) = \tau_i + \ell R_i^2 + g_i^T y_0.$$

For $w = u_i + y_0 - u_j$, include the value exactly when

$$\|(I - P_\ell)w\|^2 + (\|P_\ell w\| - s)^2 \leq R_j^2. \tag{44}$$

One witness chooses $z = -sP_\ell w / \|P_\ell w\|$ when the denominator is nonzero; otherwise any length-s kernel vector works. Every compatible singular multiplier is also a root of (43), so these families add no values beyond its six possible distinct real roots.

Candidate 14: the common circle. For the physical lens define

$$h = 1 - \frac{1 - \|a\|^2}{2p}, \qquad z = \frac{h}{\|a\|^2}a, \qquad r = \sqrt{1 - \frac{h^2}{\|a\|^2}} > 0.$$

Choose $U \in \mathbb{R}^{3\times 2}$ with orthonormal columns spanning $a^\perp$, and put

$$H_2 = U^T H U, \qquad g_2 = U^T(g + Hz), \qquad \tau_2 = q(z), \qquad \nu_2 = \operatorname{rmax}\begin{pmatrix} H_2 & I_2 \\ g_2 g_2^T/r^2 & H_2 \end{pmatrix}.$$

Always include

$$V_{\text{circ}} = \tau_2 + \nu_2 r^2 + g_2^T(\nu_2 I_2 - H_2)^\dagger g_2. \tag{45}$$

There is *no* clamp by zero in this expression: a sphere equality allows a negative multiplier. Its maximizing points are $z + U(y_0 + v)$ where $y_0 = (\nu_2 I_2 - H_2)^\dagger g_2$, $v \in \ker(\nu_2 I_2 - H_2)$ and $\|y_0 + v\| = r$.

Proof of Theorem 14. The stationary equation inside both balls is $Hx = -g$. Its complete solution set is $x_s + K$, on which $q = V_s$. Orthogonal projection turns the two original ball constraints into the two balls $B_K(k_i, \sqrt{s_i^2})$ in K. They meet exactly under (42); the displayed witness proves sufficiency.

On a positive-radius sphere the constraint gradient never vanishes. Lagrange multipliers therefore give $(\ell I - H)(x - u_i) = g_i$. Off the spectrum, the norm equation is precisely (43). At a spectral value, orthogonality to the kernel is necessary and sufficient for solvability; the pseudoinverse solution is orthogonal to the kernel. Substitution into the quadratic gives the stated constant family value. Minimizing distance to the other centre over this kernel sphere is elementary: decompose w orthogonally and choose the kernel vector opposite $P_\ell w$. The resulting squared distance is exactly the left side of (44). A compatible singular multiplier is a root because a nonzero vector in its kernel is killed by $(\ell I - H)^2 - g_i g_i^T / R_i^2$.

Subtracting the two sphere equations gives $a \cdot x = h$. Their intersection is the stated circle. Restriction to its affine plane yields the two-dimensional quadratic. The greatest-real-root proof from the main theorem applies to a sphere without imposing a nonnegative multiplier: $\nu_2 \geq \lambda_{\max}(H_2)$ still makes its matrix gap positive semidefinite, and the radial term vanishes identically. This proves (45) and its maximizers, including hard cases.

Finally the lens is compact. A global maximizer is either inside both balls, on exactly one sphere, or on their common circle. The first case is stationary; the second is a local extremum on that sphere; the third is a global extremum on the circle. The three constructions exhaust these possibilities. Conversely every included value has an explicit feasible witness. The largest candidate is therefore exactly the global maximum. All maximizers are the feasible members of precisely those families whose scalar value equals that maximum. The count is $1 + 6 + 6 + 1 = 14$. □

Proposition 15 (A physical nonglobal sphere maximum is indispensable)**.** *Take the full-rank physical Bloch data*

$$a = (-15/17, 0, 0), \quad b = (-14/85, 0, 0), \quad T = \operatorname{diag}(18/85, 2/85, 4/425), \qquad p = 10/51.$$

The unique lens optimizer is $x_* = (-2/5, 0, 0)$ *and*

$$V(p) = \frac{17509}{250000}. \tag{46}$$

It is a strict local maximum on the second boundary sphere, but not that sphere's global maximum. Keeping only the largest secular root on each sphere and the common circle gives an incorrect answer.

Proof. The canonical data are $c = (1/10, 0, 0)$ and $S = \operatorname{diag}(3/10, 1/20, 1/50)$. They are strictly positive since $\|c\| + \sum_i |S_{ii}| = 47/100 < 1$. Inverse whitening with the stated a gives the displayed rational physical data, using $\sqrt{1 - \|a\|^2} = 8/17$. The second ball has centre $(-9/2, 0, 0)$ and radius $41/10$, and

$$q(x) = \frac{929}{10000} + \frac{3}{50}x_1 + \frac{71}{10000}x_1^2 - \frac{201}{2500}x_2^2 - \frac{33}{400}x_3^2.$$

Every feasible x has $-1 \leq x_1 \leq -2/5$. Setting $x_2 = x_3 = 0$ increases q, and the remaining function has derivative at least $229/5000 > 0$ on that interval. This proves uniqueness and (46).

On the second sphere the translated gradient is $g_1 = (-39/20000, 0, 0)$ and the multiplier at x_* is $\ell = 679/102500 > 0$. Its tangent Hessian is negative definite. The opposite pole $(-43/5, 0, 0)$ gives the global sphere maximum and is infeasible for the unit ball. The unit-sphere global maximum $(1, 0, 0)$ is infeasible for the second ball. The unconstrained stationary point has $x_1 = -300/71$, and the common-circle maximum is only $118/28125 < 17509/250000$. All claims are also checked with exact rational arithmetic in the executable verification. □

13 The highest success rate preserving the unrestricted optimum

The rate of an arbitrarily chosen optimal filter need not be the best rate within a degenerate optimal family. The following formula solves that secondary problem without optimization.

Theorem 16 (Best rate and exact plateau endpoint)**.** *Assume* $\|a\| < 1$. *Let* $A = \eta I - C$, $x_0 = A^\dagger g$, $\lambda = \eta - d$ *as in the main theorem. Put*

$$K = \ker A, \quad a_K = P_K a, \quad a_R = a - a_K, \quad r = \sqrt{1 - \|x_0\|^2}, \quad \alpha = \|a_K\|, \quad u = a \cdot x_0,$$

and define

$$p_b = \frac{1 - \|a\|^2}{2(1 - u - r\alpha)}. \tag{47}$$

If $\lambda > 0$ *or* $r = 0$, *the greatest rate of an unrestricted optimal filter in the closed filter domain is* $p_{\text{best}} = p_b$. *If* $\lambda = 0$, $r > 0$ *and* $p_b r \le \alpha$, *the same formula holds. In all other cases,*

$$\boxed{p_{\text{best}} = p_i = \frac{1 - \|a_R\|^2}{1 - u + \sqrt{(1-u)^2 - r^2(1 - \|a_R\|^2)}}.} \tag{48}$$

For the boundary branch choose $x = x_0 + r a_K/\alpha$ *when* $\alpha > 0$; *any available length-*r *kernel direction works when* $\alpha = 0$. *For the interior branch choose* $x = x_0 + a_K/p_i$. *Moreover the entire probability frontier has the exact plateau*

$$V(p) = \mathcal{H}_A(\rho)^2 \quad \Longleftrightarrow \quad 0 < p \le p_{\text{best}}. \tag{49}$$

If the filtered marginal is pure instead, the frontier is constant and $p_{\text{best}} = 1$.

Proof. The unrestricted optimizing family is $x_0 + z$ with $z \in K$, $\|z\| \le r$, and equality required if $\lambda > 0$. On its boundary the rate (31) becomes $\delta/[2(1 - a \cdot x)]$. Maximizing the linear functional $a \cdot x$ on this kernel sphere gives $u + r\alpha$, proving (47). The denominator is strictly positive because $u + r\alpha \le \|a\| < 1$.

When the full disk is permitted, probability p is feasible on it exactly when the minimum squared distance to a obeys

$$\|a_R - p x_0\|^2 + \max\{\alpha - pr, 0\}^2 \le (1 - p)^2. \tag{50}$$

This formula is the orthogonal decomposition of the distance to the radius-pr kernel ball. For $pr \le \alpha$, expansion gives precisely $p \le p_b$. If $p_b r < \alpha$, the first infeasible probabilities therefore occur immediately beyond p_b. At the threshold $p_b r = \alpha$, the left side of (50) minus its right side has right derivative $2(1 - u - r\alpha) > 0$ at p_b, so infeasibility likewise begins immediately after p_b. Feasible probabilities form an interval, since a physical filter can always be scaled down. Thus no later feasible interval can occur.

If $p_b r > \alpha$, feasibility reaches $pr = \alpha$. For $pr \ge \alpha$, (50) becomes

$$-r^2 p^2 + 2(1 - u)p - (1 - \|a_R\|^2) \le 0.$$

Its smaller positive root, after rationalization, is (48). The larger root is at least one, since at $p = 1$ the left side equals $\|a_R - x_0\|^2 \ge 0$. If the larger root equals one, then $a_R = x_0$ and both roots equal one, so no isolated feasible endpoint is omitted. The smaller root is consequently the last feasible probability in $[0, 1]$. It lies beyond α/r, so the stated kernel correction is interior and attains the rate. The radicand is nonnegative: Cauchy–Schwarz bounds it below by $(\|a_R\| - \|x_0\|)^2$.

Finally $V(p)$ equals the unrestricted value precisely when the compact lens intersects the compact unrestricted maximizing family. The largest rate for such an intersection is exactly p_{best}, proving (49) and strict loss above it. For a pure filtered marginal, the product-state reduction makes every successful filtered value the same, and the identity attains it at rate one. □

The invertible convention. If no invertible filter attains the unrestricted F_3 value, the set of exact invertible optimizers is empty; a supremum over that empty set must not be substituted for p_{best}. In this case p_{best} is the greatest limiting success rate of *asymptotically* optimal invertible filters. To prove this, approach a best-rate boundary point from the ball interior; conversely any sequence of asymptotically optimal effects has a subsequence converging to the closed optimal family, and $p_{\max}(x)$ is continuous. If $\lambda = 0$ and $r > 0$, invertible exact optimizers exist, and the formula is their rate supremum, attained exactly when some best-rate point is interior. Strict violation always falls in the attained interior case.

14 Continuity of the probability frontier

Proposition 17 (Monotonicity and an explicit modulus)**.** *For a mixed filtered marginal, $V(p)$ is nonincreasing and continuous on $(0, 1]$. For $0 < p < p' \leq 1$,*

$$0 \leq V(p) - V(p') \leq 2(\|g\| + \|H\|_{\text{op}})(1 + \|a\|)\frac{p' - p}{pp'}. \tag{51}$$

It is constant near zero, with the exact plateau endpoint from Theorem 16.

Proof. Feasibility is preserved when a filter is scaled down, so $L_{p'} \subseteq L_p$. For $x \in L_p$, let

$$s = \frac{p(1 - p')}{p'(1 - p)}, \qquad x' = sx + (1 - s)a.$$

Then $0 \leq s \leq 1$, $\|x'\| \leq 1$, and $p'x' - a = [(1 - p')/(1 - p)](px - a)$, hence $x' \in L_{p'}$. Also

$$\|x - a\| \leq \frac{(1 - p)(1 + \|a\|)}{p}, \qquad \|x' - x\| \leq (1 + \|a\|)\frac{p' - p}{pp'}.$$

On the unit ball $\|\nabla q\| \leq 2(\|g\| + \|H\|_{\text{op}})$. Apply this Lipschitz bound to a maximizing $x \in L_p$ and its feasible x', and use monotonicity. The endpoint $p' = 1$ is included. □

Theorem 18 (A fixed positive rate removes every state discontinuity)**.** *For every fixed $p \in (0, 1]$, the map $\rho \mapsto \mathcal{H}_A(\rho; p)$ is continuous on the entire compact set of two-qubit density matrices, including all pure-marginal states. The same holds for Bob.*

Proof. Use all complex contractions K, a compact set in finite dimension. The admissibility condition $\text{tr}(K\rho_A K^\dagger) \geq p$ is closed, and the normalized output is continuous whenever that denominator is at least $p > 0$. If $\rho_n \to \rho$, any subsequential limit of maximizing contractions is feasible for ρ; taking limits proves upper semicontinuity.

For lower semicontinuity when $p < 1$, choose an optimal $K = U\sqrt{E}$ for ρ. If its success is strictly larger than p, it remains feasible in a neighbourhood. If the success equals p, use $E_t = (1 - t)E + tI$ and $K_t = U\sqrt{E_t}$. The success is $p + t(1 - p) > p$, while the filtered value tends to the optimal value as $t \downarrow 0$, by continuity of the positive matrix square root. Each K_t remains feasible for all sufficiently nearby states. This proves lower semicontinuity.

At $p = 1$, positivity and $\text{tr}[(I - K^\dagger K)\rho_A] = 0$ imply that K is an isometry on the support of ρ_A. The joint state is supported there, and this isometry extends to a qubit unitary. Thus every admissible output differs from ρ only by a local unitary, and $\mathcal{H}_A(\rho; 1) = F_3(\rho)$ is continuous. □

15 An elementary exact frontier for every pure two-qubit state

The following specialization connects the general formula to the familiar pure-state entanglement concentration problem [11]. It is included as a derived check and useful closed expression, not as a claim to have introduced optimal pure-state concentration.

Theorem 19 (Pure-state rate frontier)**.** *Let the smaller squared Schmidt coefficient be $\varepsilon \in [0, 1/2]$. If $\varepsilon = 0$, both one-sided frontiers equal one. If $0 < \varepsilon \le 1/2$, then for either side*

$$\boxed{\mathcal{H}_W(\rho; p)^2 = \begin{cases} 3, & 0 < p \le 2\varepsilon, \\ 1 + \dfrac{8\varepsilon(p-\varepsilon)}{p^2}, & 2\varepsilon \le p \le 1. \end{cases}} \tag{52}$$

All values are attained by invertible filters when $\varepsilon > 0$.

Proof. In Schmidt coordinates the state is $\sqrt{1-\varepsilon}|00\rangle + \sqrt{\varepsilon}|11\rangle$. Rescale a filter to have success exactly p. The unnormalized Bob marginal B obeys $0 \preceq B \preceq D = \mathrm{diag}(1-\varepsilon, \varepsilon)$ and $\mathrm{tr}\, B = p$. Writing its smaller eigenvalue as s, we have $s \le \min\{\varepsilon, p/2\}$, and $\det B = s(p-s)$ is maximized by this largest allowed s. A pure two-qubit state's correlation singular values are $1, C, C$, where $C = 2\sqrt{\det(B/p)}$. Hence $F_3^2 = 1 + 8 \det B/p^2$, giving both upper bounds in (52).

For $p \ge 2\varepsilon$, the positive invertible contraction

$$K = \mathrm{diag}\left(\sqrt{\frac{p-\varepsilon}{1-\varepsilon}}, 1\right)$$

attains the bound. At $p = 2\varepsilon$ it produces a Bell state. Scaling that filter down gives the same Bell state with any smaller positive probability. The product-state endpoint is immediate. □

For fixed $p > 0$, (52) tends to one as $\varepsilon \downarrow 0$, whereas the unrestricted squared optimum is three for every $\varepsilon > 0$. This gives an exact example of the regularization mechanism in Theorem 18.

16 Convex geometry and the complete discontinuity locus

Theorem 20 (Quasiconvexity, but not convexity)**.** *For any two density matrices and $0 \le t \le 1$,*

$$\mathcal{H}_A(t\rho + (1-t)\sigma) \le \max\{\mathcal{H}_A(\rho), \mathcal{H}_A(\sigma)\}. \tag{53}$$

Every sublevel set is closed and convex. The function itself is not convex.

Proof. For a fixed invertible filter, the filtered mixture is a mixture of the two filtered states with weights proportional to their positive success probabilities. The map F_3 is convex because it is a norm of a linear function of the state. This gives the maximum bound for that filter; taking the supremum gives (53).

More explicitly, for any fixed local filter K let $N_K(\rho)$ be the nine unnormalized correlation coordinates and let $P_K(\rho)$ be its success probability. Both are real-linear in ρ. For $v \ge 0$,

$$\{\rho : \mathcal{H}_A(\rho) \le v\} = \bigcap_K \{\rho : \|N_K(\rho)\|_2 \le v P_K(\rho)\}.$$

Each condition is a closed convex norm inequality; negative sublevels are empty. Singular filters may also be included: a zero-success positive output has trace zero and is the zero matrix, making its condition $0 \le 0$.

For nonconvexity take $\rho_1 = |00\rangle\langle 00|$ and $\rho_2 = |1\rangle\langle 1| \otimes I/2$. Their Alice-filtered values are respectively one and zero, but every $t\rho_1 + (1-t)\rho_2$, $0 < t < 1$, has supremum one by postselection on Alice's $|0\rangle$. It cannot exceed one because it is separable. Indeed a product correlation matrix has Frobenius norm at most one, and convexity proves the same for mixtures of products; local filtering preserves separability. Thus the mixture value is $1 > t$. □

Theorem 21 (Exact continuity locus)**.** *The unrestricted quantity $\mathcal{H}_A$ is lower semicontinuous everywhere. It is continuous at a state if and only if that state's Alice marginal is mixed. For $\mathcal{H}_B$, replace Alice by Bob.*

Proof. For every fixed invertible K, the success denominator is bounded below by the positive smallest eigenvalue of $K^\dagger K$, uniformly over all states. Its filtered value is therefore continuous. A supremum of continuous functions is lower semicontinuous.

At a state with $\|a\| < 1$, choose a neighbourhood where $\|a\| \le r < 1$. The fractional formula (15) is jointly continuous on that neighbourhood times the compact closed unit ball, with denominator bounded below by $(1-r)^2$. Its maximum is continuous, for example by uniform continuity and $|\max f - \max g| \le \max |f - g|$.

At any pure-Alice-marginal state, use local unitaries to write $\rho_0 = |0\rangle\langle 0| \otimes \sigma_B$ with Bloch vector b. If $\|b\| < 1$, the separable states

$$\rho_\varepsilon = (1-\varepsilon)|0\rangle\langle 0| \otimes \sigma_B + \varepsilon |1\rangle\langle 1| \otimes |\phi\rangle\langle\phi|$$

converge to ρ_0 and have $\mathcal{H}_A = 1$, whereas $\mathcal{H}_A(\rho_0) = \|b\| < 1$. If $\|b\| = 1$, locally rotate to $|00\rangle$ and approach it by the entangled pure Schmidt states of Theorem 19. Their unrestricted value is $\sqrt{3}$, whereas the product state's value is one. Every pure-marginal point is therefore a discontinuity point, completing the classification. □

17 Quantitative optimality and numerical stability

Proposition 22 (Distance and conditioning from the exact gap)**.** *Let $V = \mathcal{H}_A^2$ and retain A, x_0, λ from the main theorem. If $A \succeq mI$ with $m > 0$, then every feasible x satisfies*

$$V - q(x) \ge m \|x - x_0\|^2 . \tag{54}$$

Thus an ϵ-optimal effect lies within $\sqrt{\epsilon/m}$ of the unique optimal effect. If A is singular and nonzero, the analogous bound uses its smallest positive eigenvalue and distance to $x_0 + \ker A$. If $A = 0$, that distance is identically zero, so only the trivial distance bound applies. If $\lambda > 0$ and $V - q(x) \le \epsilon$, then

$$1 - \|x\|^2 \le \frac{\epsilon}{\lambda}, \qquad \frac{1 - \|x\|}{1 + \|x\|} \le \frac{\epsilon}{\lambda}. \tag{55}$$

The last ratio is the smaller-to-larger eigenvalue ratio of the canonical effect $I + x \cdot \sigma$.

Proof. Both statements follow from $V - q(x) = \lambda(1 - \|x\|^2) + (x - x_0)^T A (x - x_0)$. For singular A use its spectral decomposition on $(\ker A)^\perp$. For the eigenvalue ratio observe $(1-r)/(1+r) = (1-r^2)/(1+r)^2 \le 1 - r^2$ when $0 \le r \le 1$. □

Proposition 23 (Stability under coefficient perturbations)**.** *For two real symmetric quadratic problems on the unit ball,*

$$|V(\tau, H, g) - V(\widetilde{\tau}, \widetilde{H}, \widetilde{g})| \le |\tau - \widetilde{\tau}| + \left\| H - \widetilde{H} \right\|_{\mathrm{op}} + 2 \|g - \widetilde{g}\| .$$

Proof. The same expression bounds the absolute difference of their objective values at every point of the ball. Taking maxima in both directions proves the result. This avoids differentiating a repeated or singular selected root. □

The continuity theorem also explains a critical implementation rule: a nearly pure marginal must never be silently treated as exactly pure. Arbitrarily weakly entangled pure states have squared unrestricted optimum three, even when ordinary floating-point arithmetic rounds their marginal purity to one. The supplied implementations conservatively reject such ambiguous cases and request higher precision. Their behavior is tested with explicit very small Schmidt weights.

18 The complete interval of attainable values

For mixed filtered marginals, not only the largest value but the entire closed-filter range is algebraically determined. For any symmetric H and vector g, let

$$\mathcal{V}(\tau, H, g) = \tau + \mu + g^T(\mu I - H)^\dagger g, \quad \mu = \max\left\{0, \operatorname{rmax}\begin{pmatrix} H & I \\ gg^T & H \end{pmatrix}\right\}.$$

Corollary 24 (Exact attainable interval)**.** *For the canonical quadratic, the complete set of squared F_3 values under positive-success closed filters is*

$$[-\mathcal{V}(-\tau, -H, -g),\ \mathcal{V}(\tau, H, g)]\,.$$

Its endpoints have the same explicit pseudoinverse/kernel classifications, using (τ, H, g) for the upper endpoint and $(-\tau, -H, -g)$ for the lower. Every value strictly between the endpoints is attained by an invertible filter. An endpoint belongs to the invertible range exactly when its corresponding optimal family meets the open ball.

Proof. Negating the quadratic gives its minimum by the same maximum theorem. The closed ball is connected and compact, so its continuous real-valued image is the interval between its minimum and maximum. The open ball is connected and dense in the closed ball. Its image is an interval whose closure has these same two endpoints, so it contains every intermediate value. Endpoint attainment is exactly the already proved strict-ball criterion. Taking square roots gives the corresponding F_3 interval. □

19 A sharp conditional-purity bound

The previous continuity classification identifies every singular point. We can also determine exactly how large the jump can be at each point. The key is a sharp bound involving a single rank-one conditional state.

Theorem 25 (Sharp invariant conditional-purity bound)**.** *Suppose an Alice rank-one projection, of positive probability, prepares a normalized Bob state σ with Bloch-vector length $r \in [0, 1]$. Define*

$$\boxed{M(r)^2 = 1 + \frac{(1+3r)^2}{3 + 6r - r^2}.} \tag{56}$$

Then

$$\mathcal{H}_A(\rho) \le M(r). \tag{57}$$

The function M is strictly increasing, with $M(0) = \sqrt{4/3}$ and $M(1) = \sqrt{3}$. The bound is sharp for every r.

Proof. Rotate Alice's projection to $|0\rangle$ and Bob's conditional state to $\sigma = \operatorname{diag}(\alpha, \beta)$, where $\alpha = (1+r)/2$ and $\beta = (1-r)/2$. Write the density matrix in Alice blocks as

$$\rho = \begin{pmatrix} A & C \\ C^\dagger & B \end{pmatrix}, \qquad A = t\sigma, \quad \operatorname{tr} B = 1 - t, \quad 0 < t \le 1.$$

The two-qubit purity identity gives the exact expression

$$F_3(\rho)^2 = 8\operatorname{tr}(CC^\dagger) - 4|\operatorname{tr} C|^2 + 2\operatorname{tr}(A^2) + 2\operatorname{tr}(B^2) - 4\operatorname{tr}(AB) - (2t-1)^2. \tag{58}$$

For clarity, this follows by substituting the block expressions into $F_3^2 = 4\operatorname{tr}(\rho^2) - 1 - \|a\|^2 - \|b\|^2$: $\|a\|^2 = (2t-1)^2 + 4|\operatorname{tr} C|^2$ and $\|b\|^2 = 2\operatorname{tr}[(A+B)^2] - 1$.

Factor the positive matrix as $\rho = \binom{X}{Y}\binom{X}{Y}^\dagger$, so that $A = XX^\dagger$, $C = XY^\dagger$ and $B = YY^\dagger$. Then $\mathrm{tr}(CC^\dagger) = \mathrm{tr}[(X^\dagger X)(Y^\dagger Y)] \le \left\|X^\dagger X\right\|_{\mathrm{op}} \mathrm{tr}(Y^\dagger Y) = \|A\|_{\mathrm{op}} \,\mathrm{tr}\, B = t\alpha(1-t)$. This argument includes singular blocks without taking inverses. Also $\mathrm{tr}(B^2) \le (1-t)^2$, $\mathrm{tr}(AB) \ge t\beta(1-t)$, and $\mathrm{tr}(A^2) = t^2(1+r^2)/2$. Applying these inequalities in (58) yields

$$\begin{aligned} F_3(\rho)^2 &\le 1 + (2+6r)t - (3+6r-r^2)t^2 \\ &= M(r)^2 - (3+6r-r^2)\left(t - \frac{1+3r}{3+6r-r^2}\right)^2. \end{aligned} \tag{59}$$

The denominator is positive on $[0,1]$, proving the bound for the unfiltered state.

It remains to prove the asserted filtering invariance. If the initial projection vector is $|v\rangle$ and K is invertible, project the filtered state onto a normalized vector proportional to $K^{-\dagger}|v\rangle$. The resulting Bob state is the same σ, because the two factors of K cancel before normalization. It has positive probability. Thus the preceding unfiltered bound applies to every invertibly filtered state; taking its supremum proves (57).

For sharpness put $t_* = (1+3r)/(3+6r-r^2)$ and take

$$\begin{aligned} \rho_* = {} & t_*\alpha|00\rangle\langle 00| + t_*\beta|01\rangle\langle 01| + (1-t_*)|11\rangle\langle 11| \\ & + \sqrt{t_*\alpha(1-t_*)}(|00\rangle\langle 11| + |11\rangle\langle 00|). \end{aligned} \tag{60}$$

This is a positive trace-one matrix: its $00,11$ block has determinant zero and nonnegative diagonal entries, and its remaining nonzero entry is $t_*\beta \ge 0$. Every bound used above is an equality, and postselection on $|0\rangle$ gives σ. Therefore $F_3(\rho_*) = \mathcal{H}_A(\rho_*) = M(r)$. Finally direct differentiation gives

$$\frac{d}{dr}M(r)^2 = \frac{4(1+3r)(3+5r)}{(3+6r-r^2)^2} > 0,$$

proving the monotonicity and endpoint values. □

Corollary 26 (The least pure rank-one conditional state bounds the orbit)**.** *For a mixed Alice marginal define*

$$r_{\min} = \min_{\|n\|=1} \frac{\left\|b + T^T n\right\|}{1 + a\cdot n}.$$

Then $\mathcal{H}_A \le M(r_{\min})$. The minimum concerns rank-one projections, that is, the conditional surface*. Replacing it by the full steering-ellipsoid solid is invalid: even a Bell state's solid contains the origin, while all its rank-one conditional states are pure.*

Proof. The denominator is positive and the sphere is compact, so the minimum exists. Apply Theorem 25 to a minimizing projection and use monotonicity of M. □

The sharp family is an amplitude-damping or maximally-obese steering ellipsoid family after canonicalization. That family is established quantum-state geometry [5]; it is not claimed as a new family. The bound, its equality proof and the cluster-set consequence below are explicit derivations in this package. A bounded search does not establish their publication priority.

20 The exact cluster interval at every pure-marginal state

Theorem 27 (Complete singular cluster set)**.** *Fix $\rho_0 = |u\rangle\langle u| \otimes \sigma$ and let r be the Bloch length of σ. The set of limits of $\mathcal{H}_A(\rho_n)$ along convergent sequences $\rho_n \to \rho_0$ is exactly*

$$\boxed{[r, M(r)] = \left[r, \sqrt{1 + \frac{(1+3r)^2}{3+6r-r^2}}\right].} \tag{61}$$

In particular the lower limit is r, the upper limit is $M(r)$, and the maximal discontinuity jump is exactly $M(r) - r$.

Proof. By local-unitary invariance take $|u\rangle = |0\rangle$ and diagonalize σ. Lower semicontinuity already gives a lower bound r on every cluster value. For states $\rho_n \to \rho_0$, the probability of the Alice-0 projection tends to one, and its normalized Bob state tends to σ. Let its Bloch length be r_n; then $r_n \to r$ and Theorem 25 gives $\mathcal{H}_A(\rho_n) \le M(r_n) \to M(r)$.

To realize the upper endpoint use ρ_* from (60). Apply $D_\epsilon = \mathrm{diag}(1, \epsilon)$ on Alice and normalize. As $\epsilon \downarrow 0$ the state tends to $|0\rangle\langle 0| \otimes \sigma$. For each $\epsilon > 0$, the filter is invertible. The unrestricted quantity is invariant under invertible filtering on its own side, since composition permutes the group of all subsequent invertible filters. Every state in this sequence therefore has value $M(r)$.

For every intermediate value, interpolate

$$\omega_s = (1 - s)(I/2 \otimes \sigma) + s\rho_*, \qquad 0 \le s \le 1.$$

All these states have a mixed Alice marginal and the same normalized Alice-0 conditional Bob state σ. Thus $s \mapsto \mathcal{H}_A(\omega_s)$ is continuous by Theorem 21. Its endpoint values are r and $M(r)$. The intermediate value theorem supplies any prescribed value between them. Fix that s and apply the collapsing filters D_ϵ. The states converge to ρ_0, while invertible-filter invariance preserves the prescribed value for every $\epsilon > 0$. This realizes the entire interval and proves equality. □

Corollary 28 (Rigidity of the maximal one-sided value)**.** *For an arbitrary two-qubit density matrix,*

$$\mathcal{H}_A(\rho) = \sqrt{3} \quad \Longleftrightarrow \quad \rho \text{ is a pure entangled state} \quad \Longleftrightarrow \quad \mathcal{H}_B(\rho) = \sqrt{3}.$$

Proof. An entangled pure state can be converted invertibly to a Bell state by Schmidt concentration. Conversely $\sqrt{3} > 1$, so the main theorem's violating-branch result guarantees an attained invertible optimizer. Its output has $F_3^2 = 3$. The identity

$$4\,\mathrm{tr}(\rho_{\mathrm{out}}^2) = 1 + \|a_{\mathrm{out}}\|^2 + \|b_{\mathrm{out}}\|^2 + F_3(\rho_{\mathrm{out}})^2$$

and $\mathrm{tr}(\rho_{\mathrm{out}}^2) \le 1$ force zero marginal Bloch vectors and purity one, hence a Bell state up to local unitaries. An invertible filter preserves matrix rank, so the input is pure. It cannot be a product state, whose filtered value is at most one. □

Proposition 29 (An explicit global modulus at fixed success rate)**.** *For $0 < p < 1$ and $\Delta = \|\rho - \sigma\|_1$,*

$$|\mathcal{H}_A(\rho; p) - \mathcal{H}_A(\sigma; p)| \le \frac{2\sqrt{3}}{p}\left(\Delta + 2\sqrt{\frac{\Delta}{1 - p + \Delta}}\right). \tag{62}$$

At $p = 1$ the sharper bound is $\sqrt{3}\,\Delta$. The same statements hold for Bob. Thus the fixed-rate regularization is quantitatively uniform over all density matrices, without a lower bound on marginal eigenvalues.

Proof. Take a maximizing ρ-feasible filter $K = U\sqrt{E}$, and set $t = \Delta/(1-p+\Delta)$, $E_t = (1-t)E + tI$, $K_t = U\sqrt{E_t}$. Trace-norm duality gives success on σ at least $(1 - t)(p - \Delta) + t = p$. Since E_t commutes with E and their eigenvalues lie in $[0, 1]$, $\|K_t - K\|_{\mathrm{op}} \le \sqrt{t}$. The two unnormalized output matrices then differ in trace norm by at most $\Delta + 2\sqrt{t}$, using the contraction property and a two-term expansion in $K_t - K$. Normalizing matrices of trace at least p increases this bound by at most $2/p$.

Finally $|F_3(\omega) - F_3(\omega')| \le \sqrt{3}\,\|\omega - \omega'\|_1$. To verify the constant, pair the correlation difference with a Frobenius-unit real matrix R. An orthogonal singular-value decomposition gives $\left\|\sum_{ij} R_{ij}\sigma_i \otimes \sigma_j\right\|_{\mathrm{op}} \le \sum_i s_i(R) \le \sqrt{3}$; trace-norm duality and Frobenius-norm duality give the assertion. Applying it to the two normalized outputs gives one direction of (62). Interchange ρ and σ for the other. At $p = 1$, both optimized values equal the unfiltered F_3, so the last Lipschitz estimate applies directly. □

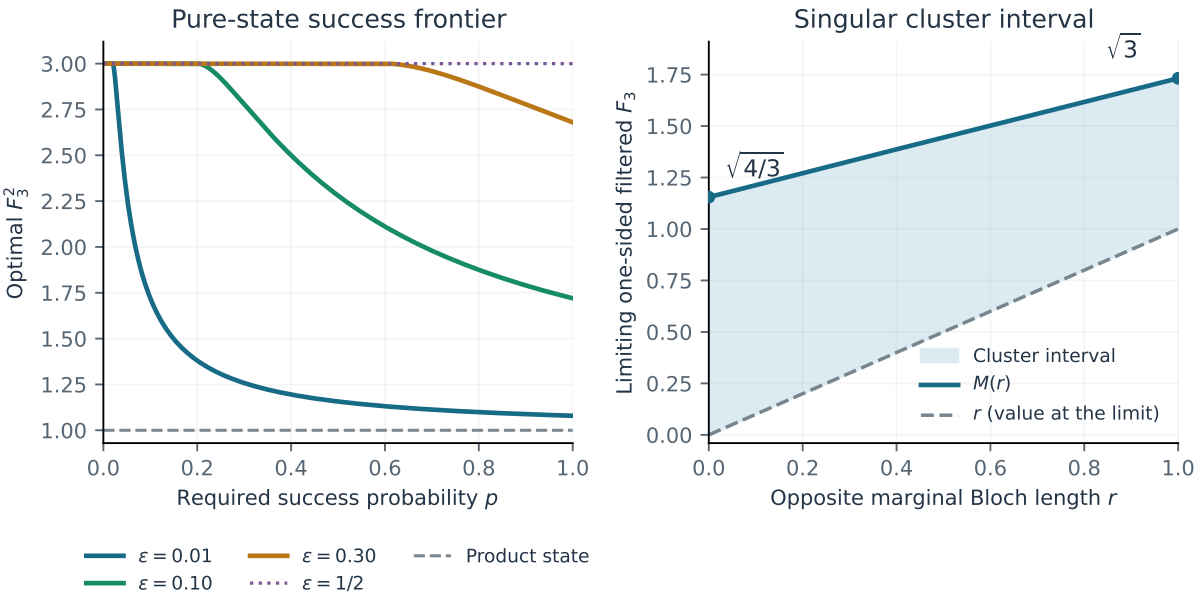


Figure 1: Exact analytical curves, without numerical filter optimization. Left: the pure-state frontier in Theorem 19; its Bell plateau ends at $p = 2\varepsilon$. Right: every value in the shaded interval is a cluster value at the corresponding pure-marginal product state, by Theorem 27. A fixed positive rate restores continuity across these singular limits. The figure is reproduced by `checks/make_figures.py`.

21 Research process and AI disclosure

This work is fully autonomous agentic research. Autonomous AI research agents produced the results, the written proofs, the exact and numerical checkers and the Lean development in an earlier session, starting from the open question of Ducuara, Susa and Skrzypczyk [1]. The records of that session describe internal audits carried out within the same automated workflow; they are internal AI-assisted reviews, not peer review, and they are not included. Opus 5.5 by Anthropic used with Claude Code assessed the package for this release: it checked the source question and the cited equations, read the proofs of Theorems 1 and 9 and of Proposition 10, reran every checker from a copy, wrote and ran the brute-force comparison, rebuilt the Lean project module by module with a new whole-project axiom audit, checked the formal statement against actual density matrices and filters, edited the manuscript and integrated the package into the repository. The mathematics of the extensions was not changed. Dev D. Goyal is responsible for publication. The work has not been externally peer reviewed. The formal scope is stated in Appendix B; the remaining results rest on the written proofs and the checks of Appendix A.

The Lean library in `lean/QuantumPrior` is adapted, with attribution, from the author's earlier formalization of steering-ellipsoid results, from which the companion package [7] was released. It contains the fully-entangled-fraction bound $f(\rho) \leq 1 - c/2$, which is Conjecture 2 of Milne, Jennings, Jevtic and Rudolph [6]. The Omega Institute's public repository records a claimed Lean proof of that conjecture committed on 8 October 2026 (issue #14206), and a proof of their Conjecture 1, on the CHSH value at fixed centre, committed on 14 October 2026 (issue #16558). No priority is claimed for either statement; neither is used in the main theorem here, and neither record contains the one-sided filtered measures.

Appendices

A Independent verification and reproducibility

All programs are in the package directory `checks/`. The command

```
python3 checks/run_checks.py --independent
```

replays them in a temporary copy with `python3 -I`.

Primary numerical implementation. `f3_exact.py` evaluates the finite spectral prescription and reconstructs valid Kraus operators. `verify.py` uses seed 20261103, generates 400 random density matrices, 100 of each rank $1, 2, 3, 4$, and checks both filtering directions (800 instances). It checks the original fractional identity, canonicalization, ellipsoid shape, reconstructed density matrices, the exact gap identity as a numerical identity, and the spectral prescription, with explicit singular examples. The maximum recovered-filter squared-value discrepancy was 9.09×10^{-13}; the largest gap-identity residual was 8.33×10^{-16}. The secondary nonsymmetric shape-spectrum check had maximum discrepancy 3.47×10^{-9}, reflecting repeated-root conditioning. These are floating-point checks and are not substituted for the proofs.

Separate implementation. `independent_check.py` was written separately. It whitens through 2×2 marginal eigendecomposition and directly filters 4×4 density matrices, rather than reusing the primary Bloch implementation. It checks 240 independently generated rank-1–4 states plus named cases (514 one-sided instances). The largest reconstruction discrepancy was 3.26×10^{-13}. For 64 instances it also compared against multistart constrained optimization, with maximum absolute discrepancy 1.70×10^{-7} and no observed optimizer excess above the formula larger than 1.78×10^{-15}. No failed checks were recorded.

Brute-force comparison. `independent_bruteforce.py` was written for this release and shares no code with the other programs. For 24 random states of ranks 1–4, some mixed with the maximally mixed state, it maximizes F_3 of the normalized Alice-filtered 4×4 state over the eight real parameters of a complex 2×2 filter by multistart Nelder–Mead and compares the result with a fresh implementation of Theorem 1. The search never exceeded the formula by more than 10^{-8} and reached it within 10^{-6} in every case. In the hard case the top eigenvalue of $\mathscr{M}$ can be defective, and a floating-point eigenvalue routine then returns a nearly real pair; the comparison therefore accepts eigenvalues with imaginary part below 10^{-6} as real. A first version with a tighter tolerance missed such a root in one case; this was an error of the comparison program, which the exact evaluators avoid.

Exact arithmetic. `verify_galois_certificate.py` uses Python's standard library, exact fractions and finite-field arithmetic. It reconstructs the sextic, verifies modular products, squarefreeness, and irreducibility, and supplies rational root/value enclosures. The finite checks support the explicitly written Galois argument; the program is not a Lean proof of field theory.

Exact evaluation for rational states. `exact_rational.py` accepts rational original Bloch data and checks all fifteen nonempty principal minors of the resulting Hermitian density matrix. It constructs the six invariant coefficients using rational arithmetic, isolates the ordered real roots of the sextic, and uses an exact Sturm count to choose the branch $\eta = \max(d, \nu)$. At a spectral pole, divisibility by the minimal polynomial determines the first nonzero derivative in Corollary 2; quotient-field reduction then certifies the returned value. No floating-point approximation chooses a branch or accepts a value. `verify_exact_rational.py` passes 23 exact cases, covering both parties, all derivative orders $0, 1, 2, 3$, the nonzero-gradient hard case, flat

optima, pure marginals, rational pure entangled states, and the degree-six Galois example. It also checks exact invariance under a rational orthogonal rotation and reconstructs determinant and adjugate identities independently. `verify_companion_exact.py` checks the companion determinant in exact integer arithmetic in dimensions 1–4, including singular shifts, and the six-invariant compression.

Operational and conditional-state checks. `extension_checks.py` checks 48 physical states at 144 probability budgets, using 1008 independent constrained-optimizer starts. The maximum reconstructed squared-value error is 1.43×10^{-13}; the maximum comparison discrepancy is 1.33×10^{-9}. It also checks 96 arbitrary affine optimal families, 72 boundary-only families against 80 directions each, and 27 pure-state frontier cases. The reference solver for the rate reports numerical success in 69 of 96 cases; all 96 endpoint values nevertheless agree within 6.14×10^{-10}. The JSON report preserves those statuses. A scalar rate expression loses 2.87×10^{-8} near a double root, while direct Hermitian effect evaluation remains accurate to 6.11×10^{-15}. `conditional_checks.py` tests 384 projections from 48 physical states, the complex block identity, invariance under invertible filters, seven exact equality families, and 21 collapsing states. The largest block-identity residual is 6.67×10^{-16} and the largest conditional-state invariance error is 2.65×10^{-15}. Five exact symbolic identities also pass. These checks detect errors; the analytical proofs are the source of the universal claims.

B Formal verification

The end-to-end theorem. The Lean declaration `FilteredF3.equation10_both_closed_forms` states, for every `M : Milne.CMatrix` (a complex matrix indexed by pairs of qubit indices) satisfying `Milne.Density M` (positive semidefinite with trace one), that

$$\mathrm{IsLUB}(\mathcal{V}_A(\rho), \sqrt{V_A(\rho)}) \quad \text{and} \quad \mathrm{IsLUB}(\mathcal{V}_B(\rho), \sqrt{V_B(\rho)}).$$

Here $\mathcal{V}_A$ is the set of values $\sqrt{\sum_{i,j} T_{ij}^2}$, with $T_{ij} = \mathrm{Re}\,\mathrm{tr}(\rho'\,\sigma_i \otimes \sigma_j)$, over the states $\rho' = (K \otimes I)\rho(K \otimes I)^\dagger/p$ for complex 2×2 matrices K with $\det K \neq 0$ and success trace $p > 0$; $\mathcal{V}_B$ is the same with $I \otimes K$. There are no rank assumptions on the state or its marginals. The right side V_B is $\|a\|^2$ when Bob's marginal is singular, and otherwise the spectral value of the canonical quadratic of the whitened state, computed from mathlib's orthonormal eigenbasis of a real symmetric 3×3 matrix and the greatest real root of the explicit secular polynomial, whose existence is proved; V_A is V_B of the swapped state. The definitions contain no optimization over filters and no assumed optimizer or certificate.

What the formal chain proves. The development proves the filtering-to-quadratic identity for actual matrices and the equality of the complete value sets, removes the canonical assumption by whitening and Pauli reconstruction, proves the product factorization at a pure filtered marginal, normalizes by the success probability and exchanges the parties by an explicit matrix swap. Invertibility of the filter is proved equivalent to the open Bloch ball, and continuity and density show that the open-ball supremum equals the closed-ball maximum. On the spectral side it proves the exact gap identity, root and optimizer existence for every finite diagonal quadratic, the transfer to arbitrary real symmetric matrices, a singular-safe block-determinant identity for the companion matrix, all maximizing coordinates and kernel freedoms, the extremizing filters and the characterization of attainment by invertible filters, and the rational violating-branch criterion in canonical coordinates. Additional modules prove scalar radius, convexity, gap-stability and parabola lemmas used by the extensions.

Not formalized. The rational Bloch-coordinate expressions for (c, Q) in Theorem 1, the direct fractional numerator, the six-invariant rule of Corollary 2, the separability corollary, the Galois obstruction, the two-ball success frontier and its candidate enumeration, the highest-success branch rule, continuity and the discontinuity locus, the pure-state frontier, the physical conditional-purity bound and its equality family, and the cluster interval are written proofs with the checks described above; for the last of these only a scalar lemma is formalized. The Lean functional is the Frobenius expression of Equation (3) of [1]; its derivation from the optimization over measurement settings is cited, not formalized. The value sets use invertible matrices up to scale; the rescaling to a contractive physical filter is in the written proof.

Build and audit. The project in `lean/` pins Lean 4.19.0 and mathlib revision `c44e0c8e e63ca166450922a373c7409c5d26b00b`. Its 43 modules keep the names of the original development; 22 of them, in `QuantumPrior`, are the attributed library described in Section 21. For this release the project was built module by module from source with `lake build`, and the module `AxiomAudit` enumerates every declaration whose defining module is one of the 41 modules that define declarations, including private and compiler-generated ones, and checks its transitive axioms with Lean's own collector. The audit covers 850 declarations, including all 490 named in the sources. The 604 declarations other than Lean 4.19 code-generation auxiliaries, among them the 503 theorems, depend only on `propext`, `Classical.choice` and `Quot.sound`; the 246 auxiliaries depend in addition only on other auxiliaries and, for 11 of them, on the code generator's placeholder `lcProof`. The fresh build compiled all 43 modules and the audit from source against the pinned mathlib cache. The sources contain no `sorry`, `admit`, `axiom` declaration, `native_decide` or `ofReduceBool`.

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